Lab Experiment-2

8-Queen Problem

Aim

To write a Python program to solve the 8-Queen problem using the Backtracking algorithm.

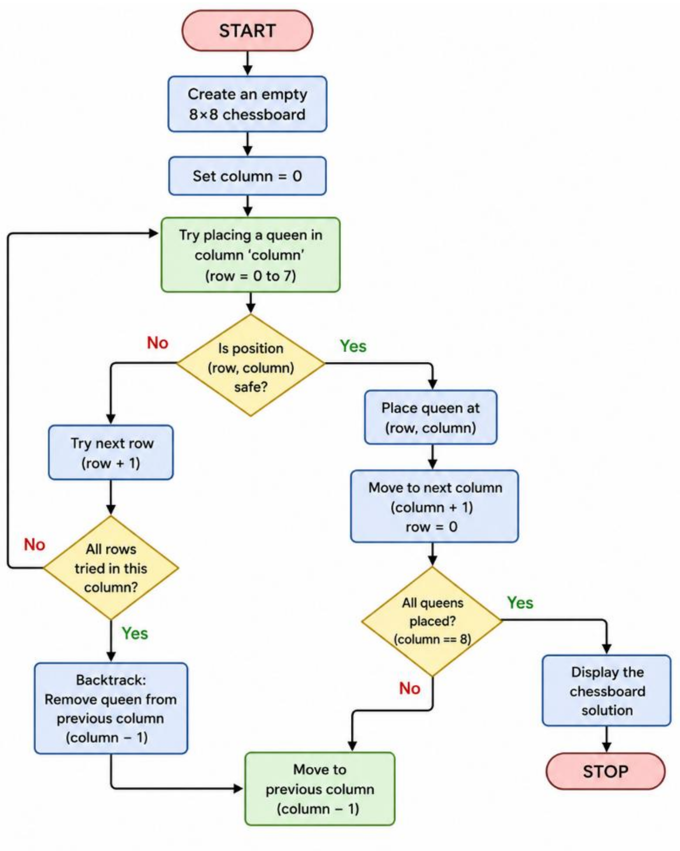
Objective

The objective of this experiment is to implement the Backtracking algorithm to solve the 8-Queen problem. It aims to place eight queens on a chessboard so that no two queens attack each other.

Algorithm

1. Create an empty 8 × 8 chessboard.
2. Start placing queens from the first column.
3. Check whether placing a queen in the current row is safe.
4. If safe, place the queen and move to the next column.
5. If no safe position is found, backtrack and try another position.
6. Repeat until all eight queens are placed successfully.
7. Display the final chessboard arrangement.

Flowchart



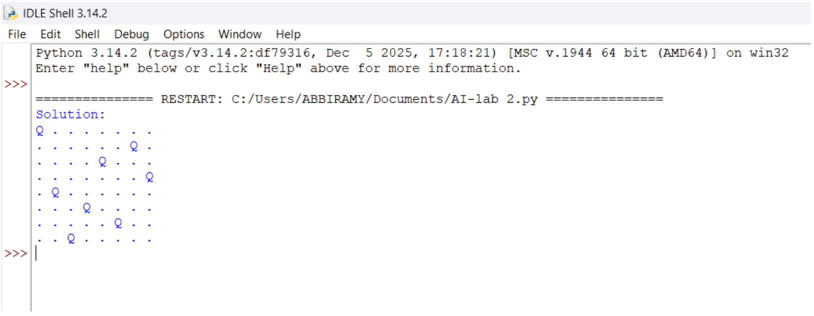
Code

```
N = 8

def print_board(board):

    for row in board:

        print(" ".join("Q" if x else "." for x in row))

def is_safe(board, row, col):

    # Check left side of row

    for i in range(col):

        if board[row][i]:

            return False

   # Check upper-left diagonal
```

```
    i, j = row, col
    while i >= 0 and j >= 0:
        if board[i][j]:
            return False
        i -= 1
        j -= 1

    # Check lower-left diagonal
    i, j = row, col
    while i < N and j >= 0:
        if board[i][j]:
            return False
        i += 1
        j -= 1
    return True
def solve(board, col):
    if col >= N:
        return True
    for row in range(N):
        if is_safe(board, row, col):
            board[row][col] = 1
            if solve(board, col + 1):
                return True
            board[row][col] = 0
    return False
board = [[0] * N for _ in range(N)]
```

```python
if solve(board, 0):
    print("Solution:")
    print_board(board)
else:
    print("No Solution Exists")
```

Output



Result

The Python program successfully solved the 8-Queen problem by placing eight queens on the chessboard without any two queens attacking each other.

Conclusion

The Backtracking algorithm efficiently finds a valid solution to the 8-Queen problem by exploring possible positions and backtracking whenever a conflict occurs.